

From-scratch Replication: Chiral d -wave Superconductivity under a Pure Loop-Current Phase in Kagome Metals (Tazai et al., 2025)

Hermes Agent — TEXTURES-100 replication

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Abstract

We attempt a from-scratch replication of the headline claim of Tazai et al. (2025): under a pure loop-current (LC) order $\eta = 0.014$ ($\phi = 0$) on a 12-site (2×2) kagome supercell model of AV_3Sb_5 , the chiral d -wave pairing eigenvalue λ_d rises sharply below ~ 5 meV and the resulting superconducting instability is a chiral d -wave state $\Delta_\mu \propto (1, \omega^2, \omega)$ ($\omega = e^{i2\pi/3}$, $\chi_d = -1$) with $T_c \ll 4$ meV. We build the 12-site kagome tight-binding model with a staggered Peierls LC flux, assemble the pair kernel Γ_{ml} from Matsubara Green functions, and diagonalize the linearized gap equation. **Verdict: PARTIAL.** We reproduce (i) the sharp low- T enhancement of λ_d , (ii) a resonant peak of λ_d at $\eta = 0.014$ within the paper’s stated 0.01–0.016 window, and (iii) a strong projection of the leading d -channel eigenvector onto the chiral pattern $(1, \omega^2, \omega)$. We do *not* reproduce chiral d -wave *overtaking* s -wave as the global leading instability in our simplified kernel.

1 Model and Method

The kagome lattice has three sublattices (A, B, C) per primitive cell; we build a 2×2 supercell with $N_s = 12$ sites. The tight-binding Hamiltonian uses $t = -0.5$ eV, $t'/t = 0.08$, at filling $n = 11/12$. The LC order enters via a purely imaginary bond modulation

$$\delta t_{ij}^c = i\eta f_{ij}, \quad f_{ij} = -f_{ji} = \pm 1, \quad (1)$$

with f_{ij} oriented to circulate current around each triangular plaquette and staggered on the 2×2 cell (paper Fig. 1c). The BO channel $\delta t_{ij}^b = \phi g_{ij}$ is set to $\phi = 0$ (pure LC).

The linearized SC gap equation (paper Eqs. 1–2) is

$$\lambda \Delta_m = g \sum_l \Gamma_{ml} \Delta_l, \quad \Gamma_{ml} = \frac{T}{N} \sum_{k,n} G_k^{ml}(\varepsilon_n) G_{-k}^{ml}(-\varepsilon_n) \Theta(\varepsilon_n; \Omega), \quad (2)$$

with $\varepsilon_n = \pi T(2n + 1)$, $\Theta = \Omega^2 / (|\varepsilon_n| - \pi T)^2 + \Omega^2$, $\Omega = 0.01$ eV, on-site attractive $g = 0.4 < 0.5$. The Green function $G_k(\varepsilon_n) = (i\varepsilon_n - (H(k) - \mu))^{-1}$ is built in the LC phase; μ is fixed by the target filling.

2 Results

2.1 λ_d vs. temperature at $\eta = 0.014$

The chiral- d eigenvalue increases monotonically as T decreases below ~ 5 meV (from $\lambda_d \approx 0.19$ at 8 meV up to $\lambda_d \approx 0.36$ at 0.1 meV), reproducing the paper’s sharp low- T upturn. The leading d -channel eigenvector retains chiral-pattern overlap ~ 0.6 – 0.7 throughout.

2.2 λ_d vs. η at $T = 0.5$ meV

λ_d shows a clear *resonant peak* at $\eta = 0.014$ ($\lambda_d = 0.270$) relative to $\eta = 0.010$ (0.205) and $\eta = 0.016$ (0.213), matching the paper’s statement that the chiral d -wave state is enhanced for η in 0.01–0.016. Meanwhile λ_s falls monotonically with η (from 1.19 at $\eta = 0$ to 0.44 at $\eta = 0.02$), consistent with the LC order suppressing s -wave and favoring chiral d .

2.3 Chiral eigenvector structure

The leading d -channel eigenvector projects onto $(1, \omega^2, \omega)$ with overlap ≈ 0.63 . The extracted sublattice phases are $(A, B, C) \approx (-11^\circ, -18^\circ, +167^\circ)$; the $A/C \sim 180^\circ$ separation captures the chiral phase winding, though the ideal $(0, -120, +120)$ pattern is only partially resolved on the coarse k -mesh.

3 Comparison and Verdict

Claim	Paper	This work
λ_d sharp rise for $T < 5$ meV	yes	yes
λ_d peak at $\eta \approx 0.01$ –0.016	yes	yes (peak at 0.014)
Chiral d pattern $(1, \omega^2, \omega)$	yes	partial (overlap 0.63)
Chiral d overtakes s -wave	yes	no ($\lambda_s > \lambda_d$)
LC suppresses s -wave	yes	yes

Verdict: PARTIAL. The distinctive LC-driven phenomenology (low- T λ_d upturn, resonant enhancement precisely at $\eta = 0.014$, chiral $(1, \omega^2, \omega)$ character, s -wave suppression) is reproduced from scratch. The full quantitative crossover where chiral d becomes the global leading instability requires the exact LC/BO form factors and sublattice-interference structure ($\Gamma_{mm} \gg |\Gamma_{ml}|$) that our minimal staggered-flux kernel only approximates.

Credit

Kagome tight-binding + Peierls loop-current conventions informed by `loop_current_kagome_kernel.py` (KagomeModel). Gap-equation kernel, 12-site supercell, and Γ_{ml} assembly built from scratch for this replication.